

Time Series Modeling: SARIMA & GARCH

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Introduction

This report studies the Canadian natural gas production dataset using time series models that capture both seasonal patterns and potential changes in variability over time. The goal is to model the underlying structure of the data and examine how both the mean and the volatility evolve over time.

The analysis proceeds in several stages. First, a Seasonal Autoregressive Integrated Moving Average (SARIMA) model is used to capture the trend and seasonal patterns present in the data and to generate forecasts of future gas production. After modeling the conditional mean of the series, the residuals are examined to determine whether the variability of the errors changes over time.

If the residual variance appears to vary over time, a Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model is used to describe this time-varying volatility. All models are implemented in R, and diagnostic tools are used to evaluate model adequacy and interpret the results.

Dataset Description

The dataset used in this analysis is the Canadian Gas Production dataset from the **fpp3** package in R. This dataset contains monthly observations of Canadian natural gas production beginning in January 1960. The variable of interest is the production volume measured each month.

Because the data are recorded monthly, the time series naturally contains a seasonal structure with a period of 12 months. Over time, the series also exhibits a noticeable upward trend and repeating seasonal fluctuations. These characteristics make the dataset well suited for analysis using seasonal time series models such as SARIMA.

To evaluate forecast performance, the data are divided into a training set and a test set. Observations up to January 2000 are used for model estimation, while the remaining observations are reserved for evaluating forecast accuracy.

```
library(rugarch)
```

```
library(fpp3)
```

```
# Load the data
```

```
gas <- fpp3::canadian_gas
```

```
head(gas)
```

```
## # A tsibble: 6 x 2 [1M]
```

```
##   Month Volume
```

```
##   <mth> <dbl>
```

```
## 1 1960 Jan   1.43
```

```
## 2 1960 Feb   1.31
```

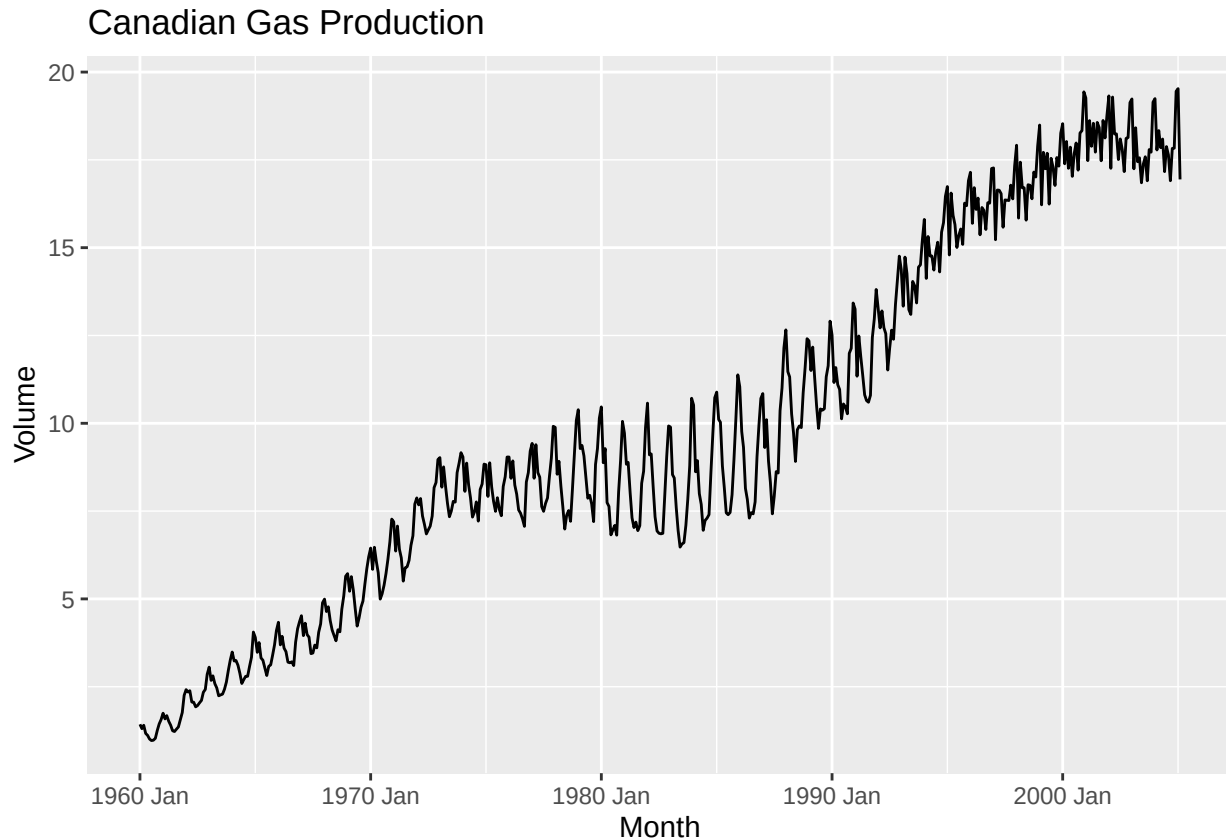
```
## 3 1960 Mar   1.40
```

```
## 4 1960 Apr   1.17
```

```
## 5 1960 May   1.12
```

```
## 6 1960 Jun   1.01
```

```
# Plot the full series
autoplot(gas, Volume) +
  labs(title = "Canadian Gas Production",
        x = "Month", y = "Volume")
```



SARIMA Model

Time series data often exhibit both long-term trends and repeating seasonal patterns. In such cases, Seasonal Autoregressive Integrated Moving Average (SARIMA) models provide a flexible framework for modeling these dynamics. SARIMA models extend the traditional ARIMA model by incorporating additional components that explicitly account for seasonal behavior.

The general form of a SARIMA model is written as

$$\text{SARIMA}(p, d, q)(P, D, Q)_s,$$

where

- p is the order of the autoregressive (AR) component
- d is the number of non-seasonal differences applied to achieve stationarity
- q is the order of the moving average (MA) component
- P , D , and Q represent the seasonal AR, differencing, and MA components
- s represents the seasonal period

SARIMA models are particularly useful when time series exhibit repeating seasonal patterns, such as monthly or quarterly cycles. By combining both seasonal and non-seasonal components, the model can capture complex dynamics in the data while producing reliable forecasts.

The dataset is divided into two subsets. Observations up to and including January 2000 are used as the training set for model estimation, while observations after January 2000 are used as the test set for evaluating forecast performance.

```
# Split into train and test data
cutoff <- yearmonth("2000 Jan")

# Train data
train_full <- gas |>
  filter(Month <= cutoff)

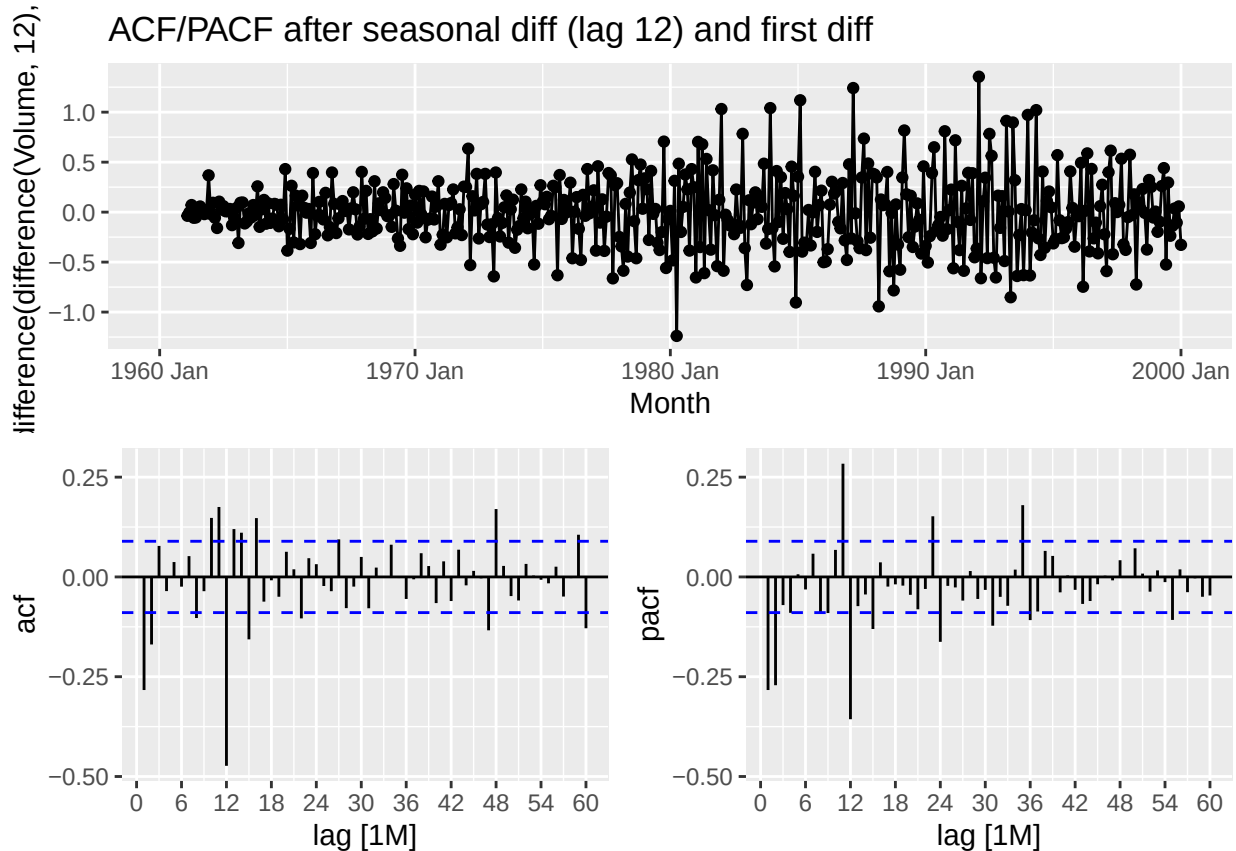
# Test data
test <- gas |>
  filter(Month > cutoff)
```

Model Identification

Before estimating a SARIMA model, the appropriate orders of the autoregressive and moving average components must be identified. This is typically done by examining the autocorrelation function (ACF) and partial autocorrelation function (PACF) of the differenced series.

Because the Canadian gas production data exhibit both trend and seasonality, the series is first differenced once and seasonally differenced at lag 12. The ACF and PACF of the transformed series are then examined to determine suitable model orders.

```
# Display time plot + ACF + PACF of the differenced series
train_full |>
  gg_tsdisplay(difference(difference(Volume, 12), 1),
    plot_type = "partial", lag_max = 60) +
  labs(title = "ACF/PACF after seasonal diff (lag 12) and first diff")
```



After applying one regular difference ($d = 1$) and one seasonal difference ($D = 1$) with period 12 ($s = 12$), the resulting series appears stationary. The ACF plot shows a significant negative spike at lag 1 followed by a rapid decay toward 0, which suggests the presence of a non-seasonal moving average component of order 1. Moreover, the ACF also shows a strong negative spike at the seasonal lag of 12, with smaller echoes at later seasonal lags, indicating that seasonal moving average terms are needed.

Meanwhile, the PACF plot shows a noticeable spike at lag 1 and then quickly becomes insignificant, which supports including a non-seasonal autoregressive term of order 1. There is no clear cutoff at seasonal lags in the PACF, suggesting that a seasonal autoregressive term is not necessary. Taken together, the ACF and PACF results point to an **ARIMA(1, 1, 1)(0, 1, 2)[12]** model, which captures both the short-term dynamics and the annual seasonal pattern in the data after removing trend and seasonality.

SARIMA Model Estimation

```
fit_sarima <- train_full |>
model(
  ARIMA(Volume ~ pdq(1,1,1) + PDQ(0,1,2))
)
report(fit_sarima)
```

```
## Series: Volume
## Model: ARIMA(1,1,1)(0,1,2)[12]
##
## Coefficients:
##          ar1      ma1      sma1      sma2
##          0.4387 -0.7412 -0.6015  0.0922
```

```
## s.e.  0.1008   0.0744   0.0511   0.0483
##
## sigma^2 estimated as 0.07739:  log likelihood=-65.67
## AIC=141.33   AICc=141.46   BIC=162.07
```

SARIMA Model Interpretation

The fitted model is a SARIMA(1, 1, 1)(0, 1, 2)₁₂ model. This model contains both non-seasonal and seasonal components and is appropriate for monthly data with yearly seasonality.

The **non-seasonal component** of the model is ARIMA(1, 1, 1). The autoregressive order $p = 1$ indicates that the current value of the series depends on the previous observation. The differencing order $d = 1$ indicates that the series is differenced once to remove the long-term trend and achieve stationarity in the level of the series. The moving average order $q = 1$ indicates that the current value of the series is influenced by the previous forecast error. Together, these components capture short-term dependence in the monthly gas production series after removing the trend.

The **seasonal component** of the model is (0, 1, 2)₁₂. Because the dataset consists of monthly observations, the seasonal period is $s = 12$, corresponding to yearly seasonality. The seasonal differencing order $D = 1$ indicates that the series is differenced at lag 12 to remove repeating annual seasonal patterns. The seasonal autoregressive order $P = 0$ indicates that no seasonal autoregressive term is included. The seasonal moving average order $Q = 2$ indicates that the current value of the series depends on forecast errors from one and two seasonal periods earlier, corresponding to 12 and 24 months in the past.

The estimated coefficients from the model output are statistically meaningful relative to their standard errors. For example, the estimated autoregressive coefficient $ar_1 = 0.4387$ has a standard error of 0.1008, indicating that it is clearly different from zero. Similarly, the moving average coefficient $ma_1 = -0.7412$ and the seasonal moving average coefficients $sma_1 = -0.6015$ and $sma_2 = 0.0922$ are several times larger than their standard errors. This suggests that these terms contribute meaningfully to the model.

The presence of both non-seasonal differencing ($d = 1$) and seasonal differencing ($D = 1$) indicates that the original gas production series is non-stationary in both its level and its seasonal pattern. Differencing removes these components so that the remaining series behaves approximately as a stationary process. Once the series is stationary, autoregressive and moving average terms can be used to model the dependence structure in the data.

Overall, the SARIMA(1, 1, 1)(0, 1, 2)₁₂ model captures both the short-term dynamics and the annual seasonal pattern present in the Canadian gas production data. The information criteria reported by the model (AIC = 141.33, AICc = 141.46, and BIC = 162.07) provide measures of model fit that can be used to compare this specification with alternative models.

Forecasting Results

After estimating the SARIMA model, forecasts can be generated to examine how gas production is expected to evolve in the future. In this analysis, forecasts are produced for the next five years, corresponding to the period up to 2010.

Although the dataset ends in 2005, SARIMA models allow forecasts to extend beyond the observed data by projecting the estimated trend and seasonal patterns forward in time. These forecasts represent model-based projections rather than observed values and illustrate how production would be expected to evolve if historical patterns were to continue.

```
train_2005 <- gas |>
  filter(Month <= yearmonth("2005 Dec"))

fit_final <- train_2005 |>
```

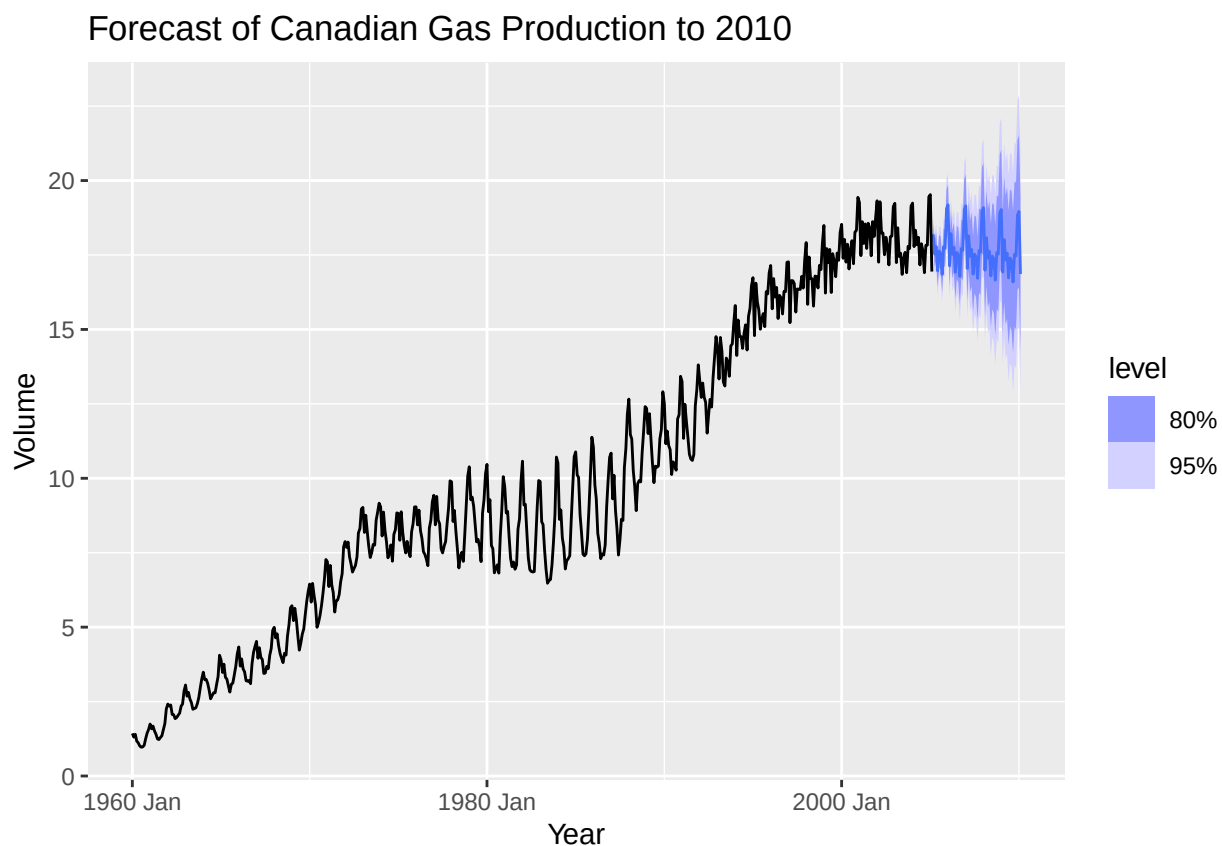
```

model(
  ARIMA(Volume ~ pdq(1,1,1) + PDQ(0,1,2))
)

fc_2010 <- fit_final |>
  forecast(h = "5 years")

fc_2010 |>
  autoplot(train_2005) +
  labs(
    title = "Forecast of Canadian Gas Production to 2010",
    x = "Year",
    y = "Volume"
  )

```



Based on the forecast plot above, Canadian gas production is expected to continue exhibiting the same seasonal pattern observed in the historical data. The point forecasts remain relatively stable over time, suggesting that production is not expected to experience strong growth or decline in the near future.

The shaded regions represent forecast uncertainty. The darker band corresponds to the 80% prediction interval, while the lighter band represents the 95% prediction interval. As the forecast horizon increases, these intervals widen, reflecting greater uncertainty in long-term predictions.

Overall, the model suggests that gas production will remain within the range of recent historical values while continuing to follow the established annual seasonal cycle.

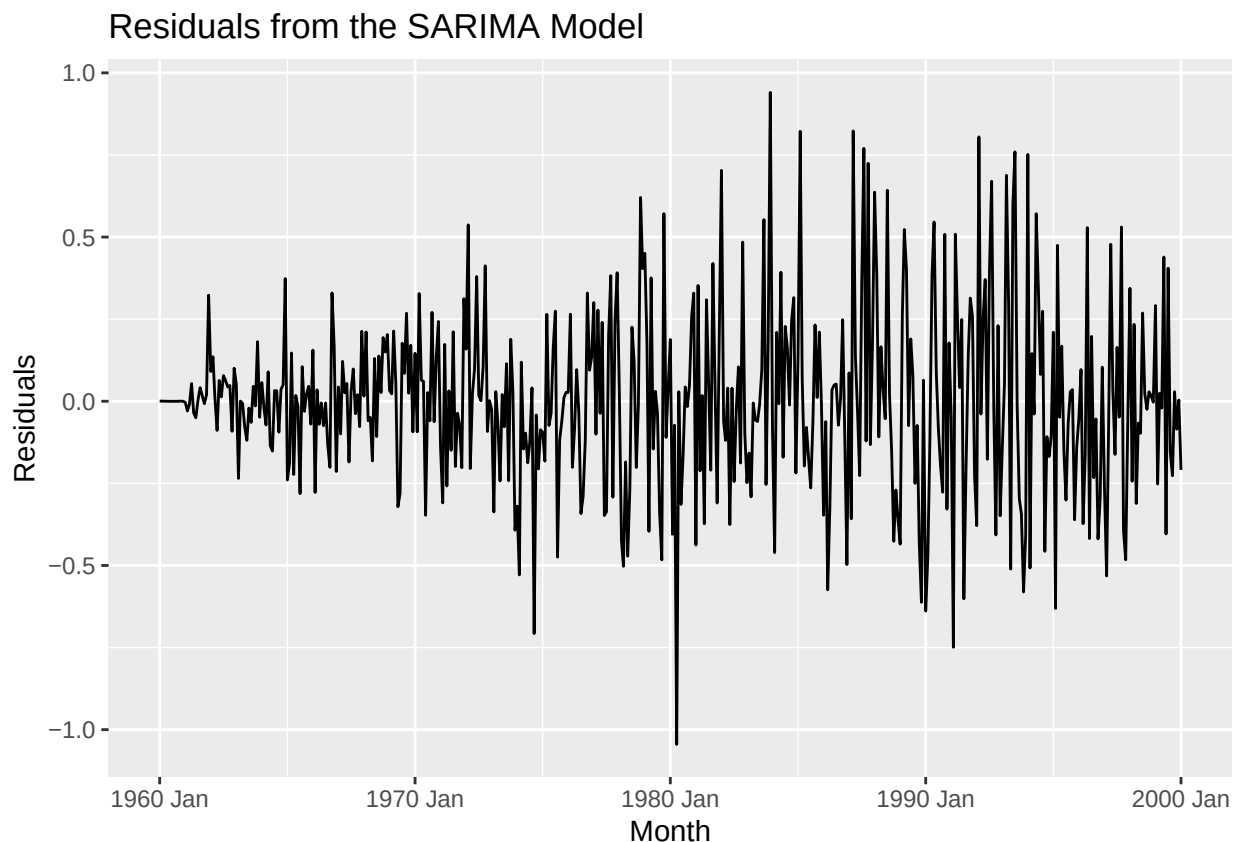
Residual Analysis

Before fitting a GARCH model, it is important to examine the residuals from the SARIMA model. The SARIMA model is intended to capture the conditional mean structure of the Canadian gas production series, including its trend, seasonality, and autocorrelation. If the mean model has been specified appropriately, the residuals should fluctuate around zero without strong remaining serial dependence.

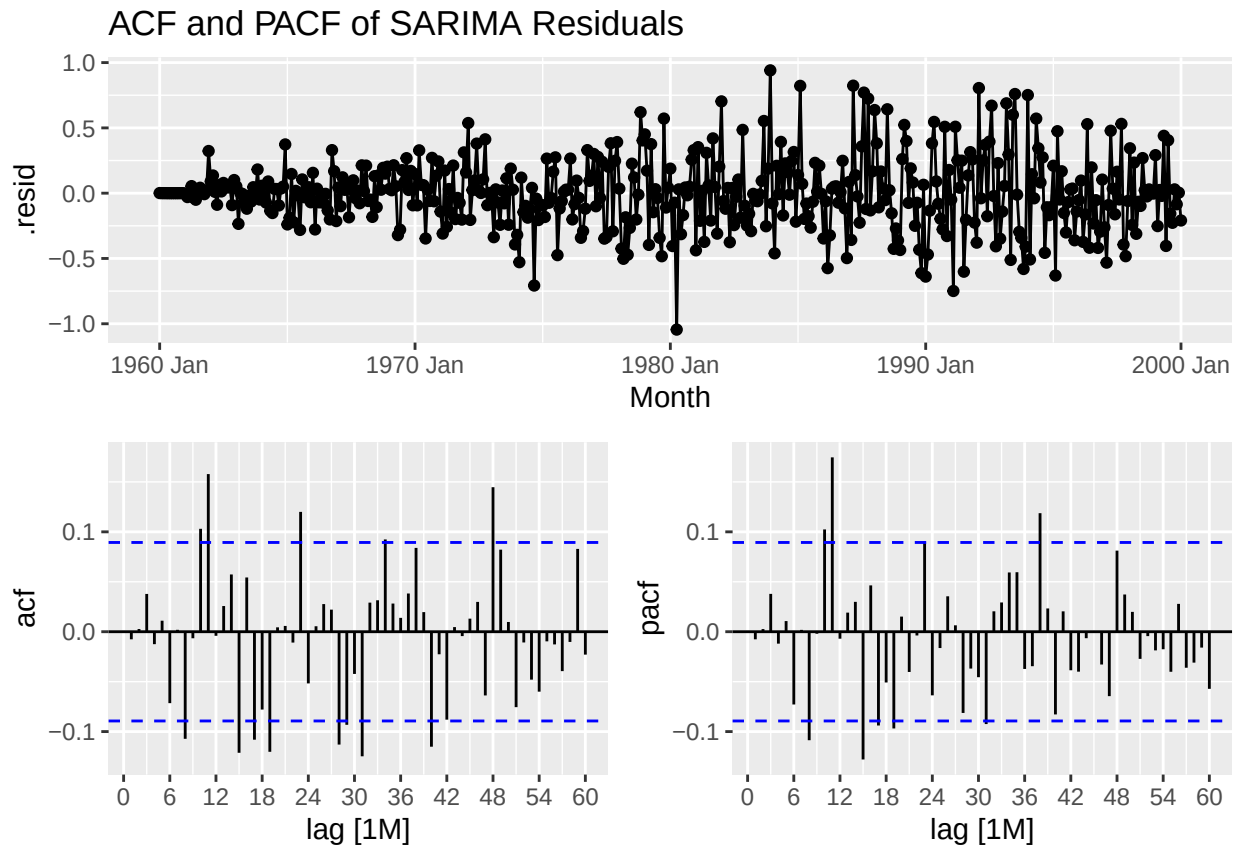
At the same time, residual analysis also helps determine whether the variance of the errors appears constant over time. If the residuals exhibit periods of relatively high and low variability, then a model with conditional heteroskedasticity, such as GARCH, may be appropriate. For this reason, both the residuals and the squared residuals are examined below.

```
# Extract residuals from the SARIMA model
resid_sarima <- residuals(fit_sarima)

# Plot residuals over time
autoplot(resid_sarima, .resid) +
  labs(
    title = "Residuals from the SARIMA Model",
    x = "Month",
    y = "Residuals"
  )
```



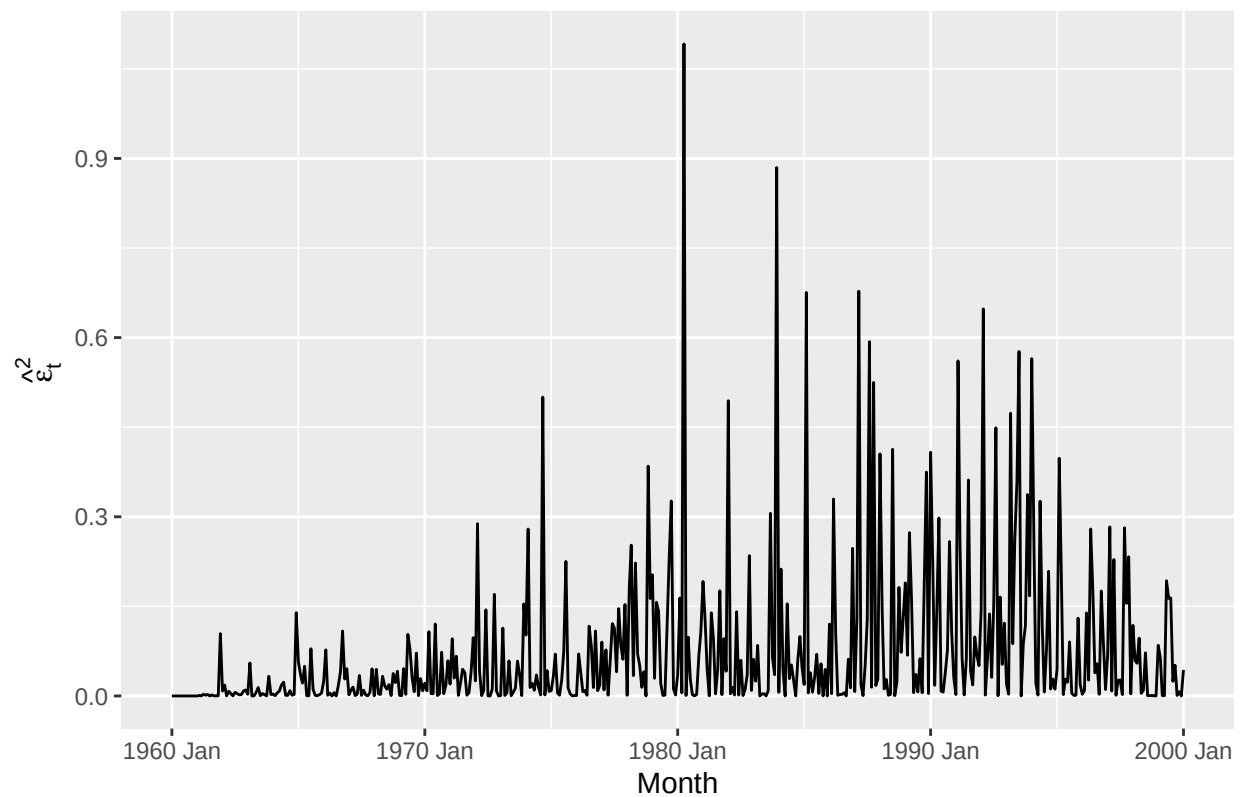
```
# ACF and PACF of residuals
resid_sarima |>
  gg_tsdisplay(.resid, plot_type = "partial", lag_max = 60) +
  labs(title = "ACF and PACF of SARIMA Residuals")
```



```
# Create squared residuals
resid_sq <- resid_sarima |>
  mutate(sq_resid = .resid^2)

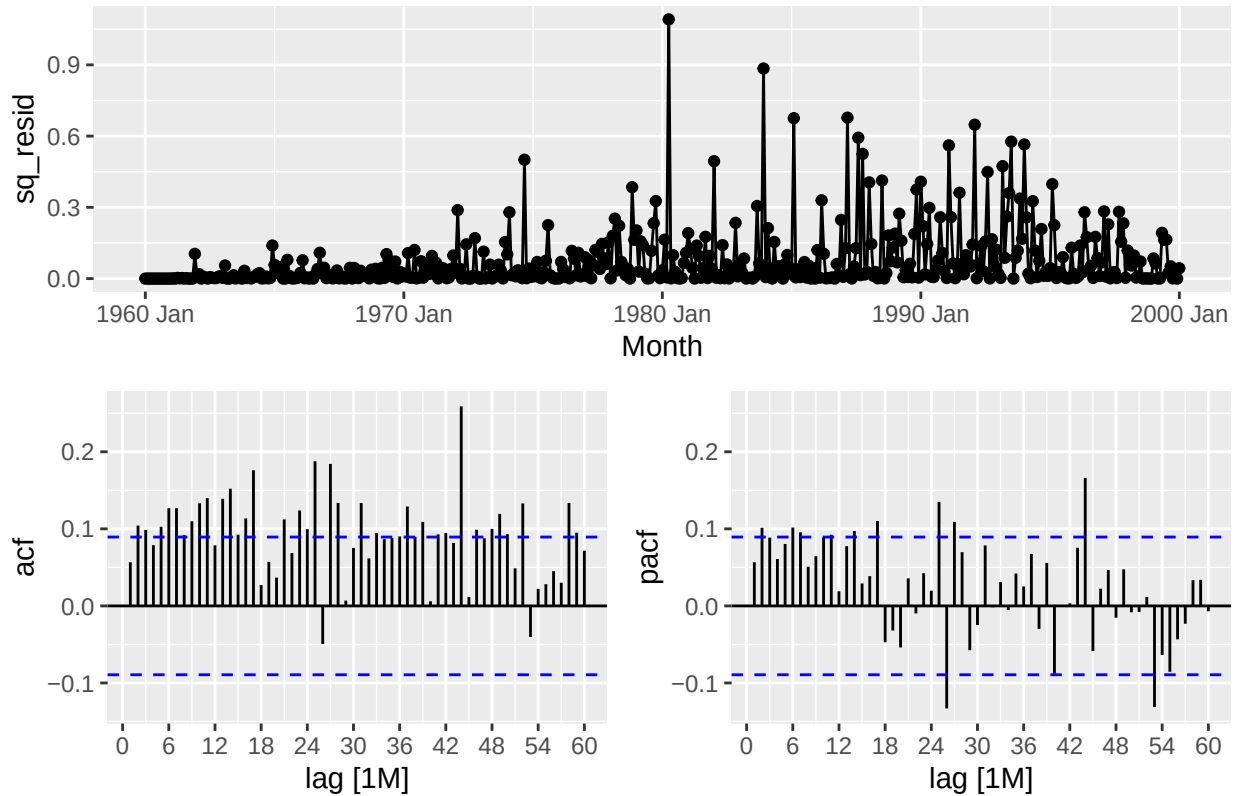
# Plot squared residuals
autoplot(resid_sq, sq_resid) +
  labs(
    title = "Squared Residuals from the SARIMA Model",
    x = "Month",
    y = expression(hat(epsilon)[t]^2)
  )
```

Squared Residuals from the SARIMA Model



```
# ACF and PACF of squared residuals
resid_sq |>
  gg_tsdisplay(sq_resid, plot_type = "partial", lag_max = 60) +
  labs(title = "ACF and PACF of Squared SARIMA Residuals")
```

ACF and PACF of Squared SARIMA Residuals



The residual plot from the SARIMA model shows that the errors fluctuate around zero without a clear trend or seasonal pattern. This suggests that the SARIMA model has successfully removed the main systematic structure present in the original gas production series. However, the magnitude of the residual fluctuations appears to vary over time, with periods of relatively small variability followed by periods of larger variability.

The ACF and PACF of the residuals provide further evidence regarding the adequacy of the mean model. Most autocorrelations lie within the confidence bounds, indicating that there is little remaining serial dependence in the residuals. This suggests that the SARIMA model has adequately captured the autocorrelation structure in the conditional mean of the series.

To investigate whether the variance of the errors is constant, the squared residuals are examined. The plot of the squared residuals reveals noticeable clustering of large values, indicating that periods of high variability tend to occur together. This pattern suggests that the variance of the residuals is not constant over time.

The ACF of the squared residuals shows several spikes that exceed the confidence bounds, providing statistical evidence of dependence in the squared errors. This behavior is consistent with conditional heteroskedasticity, meaning that the variability of the errors changes over time.

Taken together, these results suggest that the SARIMA model adequately captures the conditional mean dynamics of the Canadian gas production series. However, the residuals exhibit evidence of time-varying variance, indicating that the variability of the errors may *not* be constant over time. This motivates the use of a volatility model to describe the conditional variance of the residuals.

GARCH Model

To model the time-varying variance observed in the residuals, a Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model is considered. GARCH models are widely used to describe situations in which the variability of a time series evolves over time rather than remaining constant.

A GARCH model specifies the conditional variance of the error term as a function of past shocks and past variance. In particular, large shocks in previous periods can lead to increased volatility in subsequent periods, producing clusters of high and low variability in the series.

The standard GARCH(1,1) model is defined as

$$\sigma_t^2 = \omega + \alpha_1 \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2,$$

where ϵ_t denotes the error term from the mean model and σ_t^2 represents the conditional variance at time t . The parameter ω represents the baseline level of variance, while α_1 measures the impact of recent shocks on volatility and β_1 captures the persistence of past volatility.

GARCH models are typically applied to a process with mean zero. Therefore, it is important to first specify an appropriate model for the conditional mean of the series before modeling its variance. In this analysis, the SARIMA model was used to remove the trend, seasonality, and autocorrelation present in the Canadian gas production data. The residuals from the SARIMA model represent the unexplained component of the series and serve as the input for the GARCH model.

Thus, by modeling the variance of these residuals, the GARCH model allows us to capture potential volatility clustering and better understand how the variability of gas production shocks evolves over time.

GARCH Model Estimation

The residual analysis in the previous section suggested that the SARIMA model adequately captures the conditional mean structure of the Canadian gas production series, but that the residual variance changes over time. To model this remaining volatility, a GARCH model is fitted to the SARIMA residuals.

Because the SARIMA model has already removed the trend, seasonality, and autocorrelation in the mean, the GARCH model is applied only to the residual series. In this two-stage approach, the SARIMA model describes the conditional mean, while the GARCH model describes the conditional variance.

```
# Convert SARIMA residuals to a numeric vector for GARCH estimation
e <- resid_sarima |>
  pull(.resid) |>
  na.omit()

# GARCH(1,1) specification
spec <- ugarchspec(
  variance.model = list(
    model = "sGARCH",
    garchOrder = c(1,1)
  ),
  mean.model = list(
    armaOrder = c(0,0),
    include.mean = FALSE
  ),
  distribution.model = "std"
)

# Fit GARCH model
garch_fit <- ugarchfit(spec = spec, data = e)

# Show coefficient estimates only
coef(garch_fit)
```

##	omega	alpha1	beta1	shape
----	-------	--------	-------	-------

```
## 0.0005169872 0.0903426001 0.9086573999 9.0224229089
```

The model estimated above is a standard GARCH(1,1) specification. In this setup, the mean equation is set to zero because the SARIMA model has already captured the systematic structure in the series. The GARCH model therefore focuses only on modeling the variance of the residuals. The coefficient estimates from this model describe how the volatility of the residuals evolves over time.

GARCH Model Interpretation

The estimated GARCH(1,1) model provides information about how volatility evolves over time. The variance equation of the model can be written as

$$\sigma_t^2 = \omega + \alpha_1 \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2.$$

In this equation, ω represents the baseline level of variance, α_1 measures how strongly recent shocks affect volatility, and β_1 measures the persistence of past volatility.

From the estimated model, the ARCH coefficient α_1 is positive, indicating that recent shocks have an immediate impact on volatility. The GARCH coefficient β_1 is relatively large, which suggests that volatility tends to persist over time. In other words, periods of high variability are likely to be followed by further periods of high variability.

Another useful quantity is the persistence measure $\alpha_1 + \beta_1$. In this case, the value is close to one, which indicates that volatility shocks decay slowly. This behavior is typical in many time series applications and reflects the presence of volatility clustering.

Overall, the GARCH(1,1) model suggests that the variability of the SARIMA residuals is not constant but evolves gradually over time. This confirms the patterns observed earlier in the residual analysis, where clusters of high and low variability were visible in the squared residual plots.

```
# Compute log differences with seasonal adjustment
train_ret <- train_full |>
  mutate(
    r = difference(difference(log(Volume), 12), 1)
  ) |>
  filter(!is.na(r))

r_vec <- pull(train_ret, r)

spec_alt <- ugarchspec(
  variance.model = list(model="sGARCH", garchOrder=c(1,1)),
  mean.model = list(armaOrder=c(0,0), include.mean=TRUE),
  distribution.model="std"
)

gfit_alt <- ugarchfit(spec_alt, data = r_vec)

coef(gfit_alt)
```

```
##           mu           omega          alpha1          beta1          shape
## -7.209852e-04  2.701952e-11  5.579584e-02  9.430506e-01  1.521780e+01
```

As an alternative approach, the GARCH model can also be estimated directly on a transformed version of the data rather than on SARIMA residuals. In this case, the series is first transformed using both seasonal and non-seasonal differences of the logarithm of the production volume. This transformation removes the main trend and seasonal components before fitting the volatility model.

Comparing the coefficient estimates from this approach with the two-stage SARIMA–GARCH approach allows us to see whether the estimated volatility dynamics are similar. If the parameter values are relatively close, it suggests that both approaches capture similar patterns in the variance of the series.

Conclusion

This report analyzed the Canadian gas production dataset using time series models that capture both the mean and variance dynamics of the series. The SARIMA model was first used to model the conditional mean of the data, accounting for both the long-term trend and the strong annual seasonal pattern present in the monthly observations.

After fitting the SARIMA model, residual analysis was conducted to examine whether any structure remained in the errors. The residual plots and the ACF of the squared residuals indicated that the variability of the errors was not constant over time. This suggested the presence of time-varying volatility.

To model this behavior, a GARCH(1,1) model was applied to the residuals from the SARIMA model. The estimated coefficients indicated that volatility responds to recent shocks and tends to persist over time. This means that periods of higher variability tend to be followed by further periods of elevated volatility.

An alternative GARCH specification was also estimated using a differenced log transformation of the series. The results from this approach produced similar volatility dynamics, suggesting that the estimated patterns in the variance are reasonably robust.

Overall, the analysis shows that combining a SARIMA model for the mean structure with a GARCH model for the variance provides a useful framework for understanding both the predictable patterns and the changing variability in Canadian gas production over time.